开源软件基础

matplotlib

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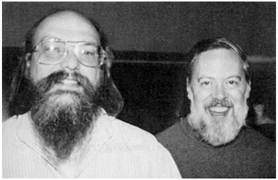
Outline

今天就能开始用的小知识

creat

Ken Thompson was once asked what he would do differently if he were redesigning the UNIX system. His reply: "I'd spell creat with an e."[a]

[a] https://en.wikiquote.org/wiki/Ken_Thompson



今天就能开始用的小知识

spell it with an "e"

Browse files

```
R=rsc
http://go/go-review/1025037
```

master   weekly.2012-03-27 … 1.10beta1.mailed

ken committed on 11 Nov 2009   1 parent 8a902ed   commit c90d392ce3d3203e0c32b3f98d1e68c4c2b4c49b

Showing **1 changed file** with **1 addition** and **0 deletions**.

Unified | Split

1 src/pkg/os/file.go   View

```
@@ -62,6 +62,7 @@ const (
62 62          O_NDELAY    = O_NONBLOCK;       // synonym for O_NONBLOCK
63 63          O_SYNC      = syscall.O_SYNC;   // open for synchronous I/O.
64 64          O_TRUNC     = syscall.O_TRUNC;  // if possible, truncate file when opened.
   65  +       O_CREATE    = O_CREAT;          // create a new file if none exists.
65 66   )
66 67
67 68   // Open opens the named file with specified flag (O_RDONLY etc.) and perm, (0666 etc.)
```

3 comments on commit c90d392

dotancohen replied on 17 Oct 2015

Ken, you waited 25 long years to make this commit!

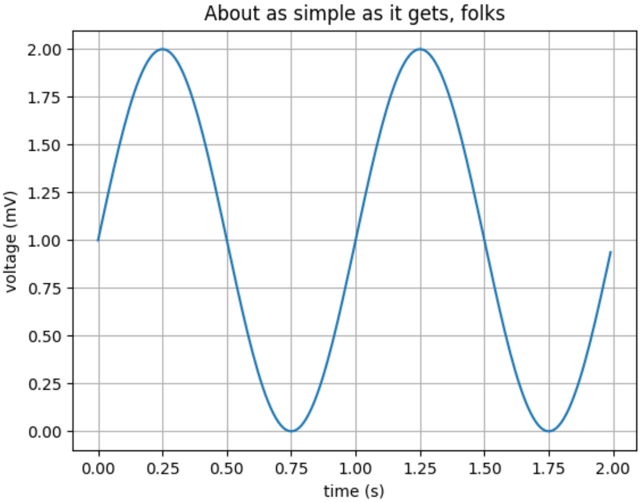
👍 12

Outline

操作步骤

- 使用 import numpy 引入 numpy 库
- 使用 numpy.linspace 或 numpy.arange 或列表准备数据
- 使用 import matplotlib.pyplot as plt 引入包
- 使用 plt.plot(x, y) 和 plot.show() 画图（推荐 jupyter）
- 使用 plt.xlabel(”)，plt.ylabel(”)，plt.title(”) 设置标题
- 使用 plt.savefig(fname, dpi) 保存图片
- 使用 fig, ax = plt.subplots() 返回图片句柄和坐标轴
- fig.savefig() 保存图片
- ax.plot 画图
- ax.set(xlabel=“x”, ylabel=“y”, title=“title”)

Simple Plot


About as simple as it gets, folks
voltage (mV)
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
time (s)

Simple Plot

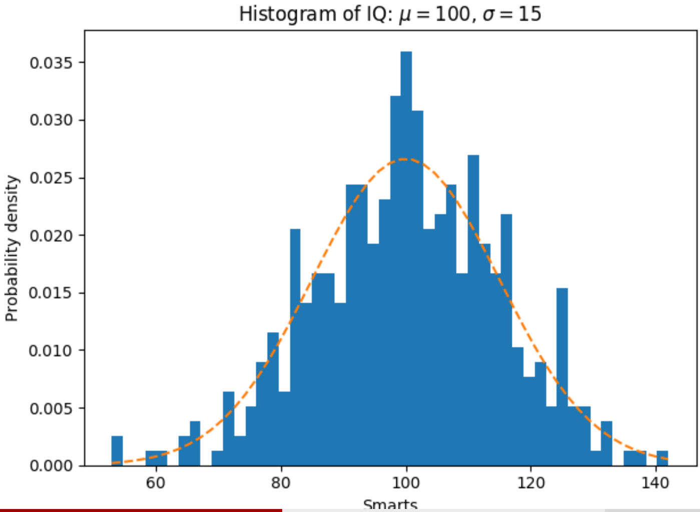
```
import matplotlib.pyplot as plt
import numpy as np

# Data for plotting
t = np.arange(0.0, 2.0, 0.01)
s = 1 + np.sin(2 * np.pi * t)

# Note that using plt.subplots below is equivalent to using
# fig = plt.figure and then ax = fig.add_subplot(111)
fig, ax = plt.subplots()
ax.plot(t, s)

ax.set(xlabel='time (s)', ylabel='voltage (mV)',
       title='About as simple as it gets, folks')
ax.grid()

fig.savefig("test.png")
plt.show()
```

Outline

Demo of the histogram (hist) function with a few features

Selecting different bin counts and sizes can significantly affect the shape of a histogram. The Astropy docs have a great section on how to select these parameters:
http://docs.astropy.org/en/stable/visualization/histogram.html

Demo of the histogram (hist) function with a few features



Demo of the histogram (hist) function with a few features

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(19680801)
mu, sigma = 100, 15
x = mu + sigma * np.random.randn(10000)

# the histogram of the data
n, bins, patches = plt.hist(x, 50, normed=1, facecolor='g', alpha=0.75)

plt.xlabel('Smarts')
plt.ylabel('Probability')
plt.title('Histogram of IQ')
plt.axis([40, 160, 0, 0.03])
plt.grid(True)
plt.show()
```

Outline

Basic pie chart

Demo of a basic pie chart plus a few additional features.

Basic pie chart

In addition to the basic pie chart, this demo shows a few optional features:

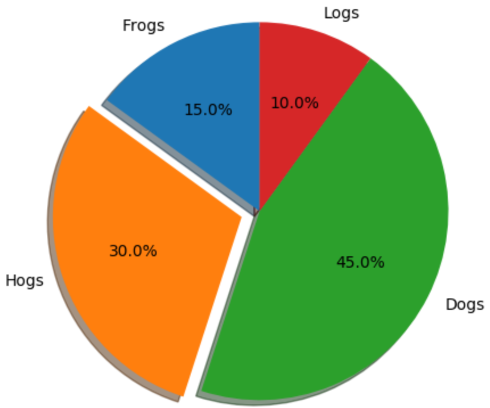
Basic pie chart

Note about the custom start angle:

Basic pie chart

The default startangle is 0, which would start the “Frogs” slice on the positive x-axis. This example sets startangle = 90 such that everything is rotated counter-clockwise by 90 degrees, and the frog slice starts on the positive y-axis.

Basic pie chart



Basic pie chart

```
import matplotlib.pyplot as plt

# Pie chart, where the slices will be ordered and plotted counter-clockwis
labels = 'Frogs', 'Hogs', 'Dogs', 'Logs'
sizes = [15, 30, 45, 10]
explode = (0, 0.1, 0, 0)  # only "explode" the 2nd slice (i.e. 'Hogs')

fig1, ax1 = plt.subplots()
ax1.pie(sizes, explode=explode, labels=labels, autopct='%1.1f%%',
        shadow=True, startangle=90)
ax1.axis('equal')  # Equal aspect ratio ensures that pie is drawn as a cir

plt.show()
```

Outline

Fill plot demo

Demo fill plot.

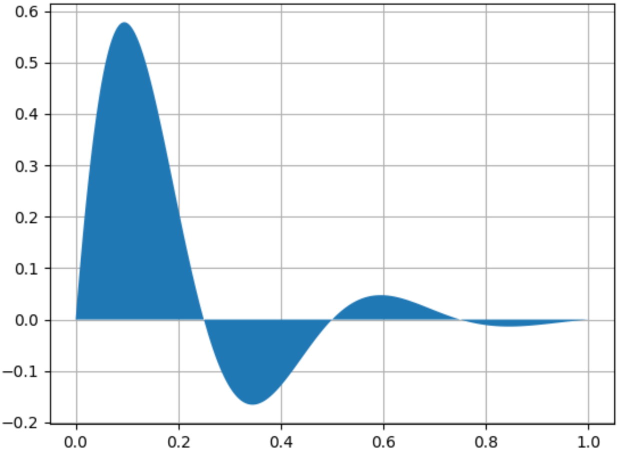
Fill plot demo

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0, 1, 500)
y = np.sin(4 * np.pi * x) * np.exp(-5 * x)
```

Fill plot demo

```
fig, ax = plt.subplots()

ax.fill(x, y, zorder=10)
ax.grid(True, zorder=5)

x = np.linspace(0, 2 * np.pi, 500)
y1 = np.sin(x)
y2 = np.sin(3 * x)
```

Fill plot demo

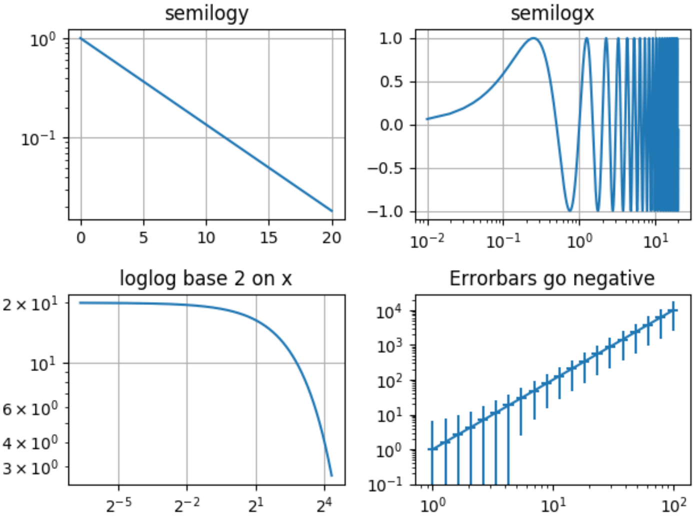


Outline

Log Demo

Examples of plots with logarithmic axes.

Log Demo



semilogy
10^0
10^{-1}
0
5
10
15
20
semilogx
1.0
0.5
0.0
−0.5
−1.0
10^{-2}
10^{-1}
10^0
10^1
loglog base 2 on x
2×10^1
10^1
6×10^0
4×10^0
3×10^0
2^{-5}
2^{-2}
2^1
2^4
Errorbars go negative
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^0
10^1
10^2

Log Demo

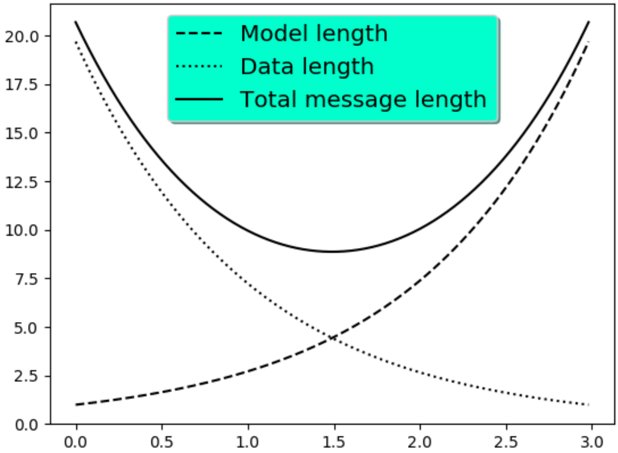
```
import numpy as np
import matplotlib.pyplot as plt

# Data for plotting
t = np.arange(0.01, 20.0, 0.01)

# Create figure
fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(2, 2)

# log y axis
ax1.semilogy(t, np.exp(-t / 5.0))
ax1.set(title='semilogy')
ax1.grid()

# log x axis
ax2.semilogx(t, np.sin(2 * np.pi * t))
ax2.set(title='semilogx')
ax2.grid()

# log x and y axis
ax3.loglog(t, 20 * np.exp(-t / 10.0), basex=2)
```

Outline

Legend using pre-defined labels

Notice how the legend labels are defined with the plots!

Legend using pre-defined labels



Legend using pre-defined labels

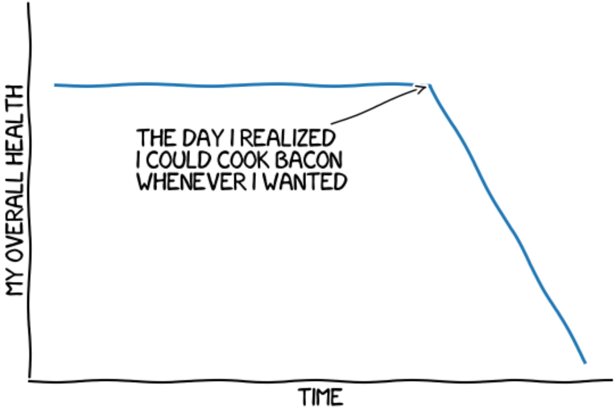
```
import numpy as np
import matplotlib.pyplot as plt

# Make some fake data.
a = b = np.arange(0, 3, .02)
c = np.exp(a)
d = c[::-1]

# Create plots with pre-defined labels.
fig, ax = plt.subplots()
ax.plot(a, c, 'k--', label='Model length')
ax.plot(a, d, 'k:', label='Data length')
ax.plot(a, c + d, 'k', label='Total message length')

legend = ax.legend(loc='upper center', shadow=True, fontsize='x-large')

# Put a nicer background color on the legend.
legend.get_frame().set_facecolor('#00FFCC')

plt.show()
```

Outline

XKCD

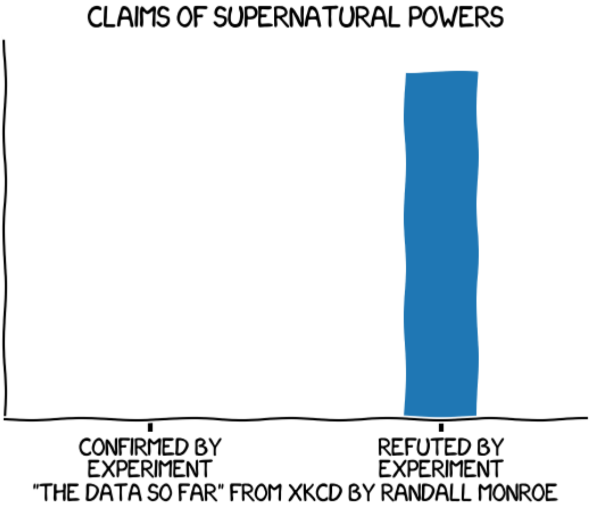
Shows how to create an xkcd-like plot.

XKCD



"STOVE OWNERSHIP" FROM XKCD BY RANDALL MONROE

XKCD


CLAIMS OF SUPERNATURAL POWERS
CONFIRMED BY EXPERIMENT
REFUTED BY EXPERIMENT
"THE DATA SO FAR" FROM XKCD BY RANDALL MONROE

XKCD

```
import matplotlib.pyplot as plt
import numpy as np

with plt.xkcd():
    # Based on "Stove Ownership" from XKCD by Randall Monroe
    # http://xkcd.com/418/

    fig = plt.figure()
    ax = fig.add_axes((0.1, 0.2, 0.8, 0.7))
    ax.spines['right'].set_color('none')
    ax.spines['top'].set_color('none')
    plt.xticks([])
    plt.yticks([])
    ax.set_ylim([-30, 10])

    data = np.ones(100)
    data[70:] -= np.arange(30)

    plt.annotate(
        'THE DAY I REALIZED\nI COULD COOK BACON\nWHENEVER I WANTED',
        xy=(70, 1), arrowprops=dict(arrowstyle='->'), xytext=(15, -10))
```